

Question:

If the parabola $y_1 = x^2 + 2x + 7$ and the line $y_2 = 6x + b$ intersect at only one point, what is the value of b ?

Answer Choices:

- A. 10
- B. 5
- C. 3
- D. 12
- E. 4
- F. 6
- G. 8
- H. 7
- I. 2
- J. 9

Reasoning + Answer:

`<think>` To find the value of b such that the parabola $y_1 = x^2 + 2x + 7$ and the line $y_2 = 6x + b$ intersect at only one point, we need to consider the discriminant of the quadratic equation formed by equating the two expressions. Since the parabola and the line intersect at only one point, the discriminant must be equal to zero. `</think>`
`<answer>` A. 10 `</answer>`

